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## Connector, cable connection assembly and connector assembly


## Abstract

A connector includes an insulator, a plurality of pairs of terminals provided on the insulator, and a resistance element. The plurality of pairs of terminals include at least one pair of cable terminals electrically connected with a cable and at least one pair of vacant terminals that are unused. The resistance element is electrically connected between each pair of vacant terminals.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the benefit of the filing date under 35 U.S.C. § 119(a)-(d) of Chinese Patent Application No. 202110913098.X, filed on Aug. 10, 2021.

FIELD OF THE INVENTION

(2) The present invention relates to a connector, a cable connection assembly including the connector, and a connector assembly including the connector or the cable connection assembly.

BACKGROUND

(3) In the prior art, it is sometimes necessary to use a cable to connect a first connector with four pairs of terminals to a second connector with two pairs of terminals. There are then two pairs of unused terminals in the first connector. In the prior art, the two pairs of vacant terminals are suspended in the first connector, like two antennas, which will generate electromagnetic radiation, interfere with the surrounding signal terminals, and affect the signal transmission.

SUMMARY

(4) A connector includes an insulator, a plurality of pairs of terminals provided on the insulator, and a resistance element. The plurality of pairs of terminals include at least one pair of cable terminals

electrically connected with a cable and at least one pair of vacant terminals that are unused. The resistance element is electrically connected between each pair of vacant terminals.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will now be described by way of example with reference to the accompanying Figures, of which:

(2) FIG. **1** shows an illustrative perspective view of a cable connection assembly according to an exemplary embodiment of the present invention;

(3) FIG. **2** shows an illustrative perspective view of a first connector in the cable connection assembly shown in FIG. **1**;

(4) FIG. **3** shows a longitudinal sectional view of the first connector shown in FIG. **2**;

(5) FIG. **4** shows a schematic diagram of the insulator, terminal, and positioning end cap of the first connector shown in FIG. **3**;

(6) FIG. **5** shows a schematic diagram of the insulator, terminal and resistance element of the first connector shown in FIG. **3**; and

(7) FIG. **6** shows a schematic diagram of a mating connector and a circuit board according to an exemplary embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(8) Exemplary embodiments of the present disclosure will be described hereinafter in detail with reference to the attached drawings, wherein like reference numerals refer to like elements. The present disclosure may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, these embodiments are provided so that the present disclosure will convey the concept of the disclosure to those skilled in the art.

(9) In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.

(10) As shown in FIGS. **1** to **3**, in the illustrated embodiment, the cable connection assembly mainly includes a first connector **10**, a second connector **20**, and a cable **30** connected between the first connector **10** and the second connector **20**.

(11) As shown in FIGS. **1** to **5**, in the illustrated embodiment, the first connector **10** includes an insulator **110** and a plurality of pairs of terminals **120**. The plurality of pairs of terminals **120** are respectively installed in the terminal slots formed on the insulator **110**. The plurality of pairs of terminals **120**, **120′** of the first connector **10** include at least one pair of cable terminals **120** for electrical connection with the cable **30** and at least one pair of unused vacant terminals **120′**. A resistance element **190** is electrically connected between each pair of vacant terminals **120′**. In the embodiment shown in this figure, each pair of vacant terminals **120′** forms an electrical circuit through the resistance element **190**, thereby avoiding the antenna effect caused by the vacant terminals **120′** and improving the reliability of signal transmission of the connector.

(12) As shown in FIGS. **1** to **5**, in the illustrated embodiment, the connector **10** includes two pairs of vacant terminals **120′** and two resistance elements **190**. The resistance element **190** is connected in series between the rear ends of each pair of vacant terminals **120′**. The connector **10** includes two pairs of cable terminals **120** for electrically connecting to two pairs of core wires **31** of the cable **30**.

(13) In the embodiment shown in FIG. **5**, each resistance element **190** includes two pins **190*a***. The rear end of the vacant terminal **120′** is crimped or welded to the pin **190*a*** of the resistance element

**190**. The front end of the vacant terminal **120′** is suitable for mating with a mating terminal **52** of the mating connector **50**.
(14) As shown in FIG. **3**, in the illustrated embodiment, the rear end of the cable terminal **120** is adapted to be crimped to the core wire **31** of the cable **30** to electrically connect with the core wire **31** of the cable **30**. The front end of the cable terminal **120** is suitable for mating with the mating terminal **52** of the mating connector **50**.
(15) In the embodiment shown in FIG. **3**, the first connector **10** also includes a housing **130**, a shield shell **140**, and a sealing ring **101**. The housing **130** is sleeved on the insulator **110**. The shield shell **140** is sleeved on the housing **130**. The seal ring **101** is compressed between the housing **130** and the shield shell **140** to seal the gap between the housing **130** and the shield shell **140**.
(16) The first connector **10** further includes a conductive screw sleeve **150** and a conductive spring **103**, as shown in FIG. **3**. The conductive screw sleeve **150** is sleeved on the housing **130** and the shield shell **140** and is suitable for threaded connection with a mating shield shell **51** of the mating connector **50**. The conductive spring **103** is compressed between the shield shell **140** and the conductive screw sleeve **150** to electrically connect the shield shell **140** and the conductive screw sleeve **150**.
(17) As shown in FIG. **3**, in the illustrated embodiment, the first connector **10** also includes a heat shrinkable tube **160**. One end of the heat shrinkable tube **160** is heat shrunk onto the shield shell **140**, and the other end is heat shrunk onto the cable **30**. The shielding layer **32** of the cable **30** is clamped between the heat shrinkable tube **160** and the shield shell **140** and welded to the shield shell **140** via a copper foil.
(18) As shown in FIG. **3**, in the illustrated embodiment, the first connector **10** also includes a sealant **102**, which is filled in the rear port of the shield shell **140** to seal the rear port of the shield shell **140**.
(19) In the embodiment shown in FIG. **3**, the core wire **31** of the cable **30** extends into the shield shell **140** from the rear port of the shield shell **140**, and the resistance element **190** is provided in the rear port of the shield shell **140**.
(20) As shown in FIG. **4**, in the illustrated embodiment, the first connector **10** also includes an isolation sleeve **191**, which is sleeved on the resistance element **190** to separate the resistance element **190** from the sealant **102**.
(21) In the embodiment shown in FIGS. **3** and **4**, the first connector **10** also includes a plurality of positioning end caps **170**. The plurality of positioning end caps **170** are assembled on the front end of the insulator **110** for positioning the front ends of the plurality of pairs of terminals **120** and **120′**, respectively. A pair of positioning holes **171** is formed in each positioning end cap **170**. The front ends of each pair of terminals **120** and **120′** pass through the pair of positioning holes **171** of the corresponding positioning end cap **170**.
(22) As shown in FIG. **2**, the cable connection assembly also includes a tail sleeve **180**, which is molded on the heat shrinkable tube **160** and the cable **30** of the first connector **10** by an insert injection molding manner. That is, the tail sleeve **180** is an insert injection molded piece.
(23) As shown in FIG. **3**, the cable **30** has a core wire **31** and a shielding layer **32**. The core wire **31** of the cable **30** is electrically connected to the cable terminal **120** of the first connector **10**, and the shielding layer **32** of the cable **30** is electrically connected to the shield shell **140** of the first connector **10**. The second connector **20** shown in FIG. **1** includes a cable terminal. The cable terminal of the second connector **20** is electrically connected to the core wire **31** of the cable **30**, so that the cable terminal **120** of the first connector **10** is electrically connected to the cable terminal of the second connector **20** through the core wire **31** of the cable **30**. In the illustrated embodiment, the second connector **20** includes two pairs of cable terminals, and the cable **30** has two pairs of core wires **31**. The two pairs of cable terminals **120** of the first connector **10** are electrically connected to the two pairs of cable terminals of the second connector **20** through the two pairs of core wires **31** of the cable **30**.

(24) As shown in FIGS. **1** to **6**, in an exemplary embodiment of the present invention, a connector assembly is also disclosed. The connector assembly includes: the first connector **10** or the cable connection assembly; and a mating connector **50** adapted to mate with the first connector **10**.
(25) As shown in FIG. **6**, the mating connector **50** includes a mating shield shell **51**, a plurality of pairs of mating terminals **52** provided in the mating shield shell **51**, and an insulation holder for holding the mating terminals **52**. When the first connector **10** and the mating connector **50** are mated together, the shield shell **140** of the first connector **10** is electrically connected to the mating shield shell **51** of the mating connector **50** through the conductive screw sleeve **150**, and the plurality of pairs of terminals **120** and **120′** of the first connector **10** are mated with the plurality of pairs of mating terminals **52** of the mating connector **50** respectively.
(26) As shown in FIG. **6**, in the illustrated embodiment, the connector assembly also includes a circuit board **40**. The mating connector **50** is mounted on the circuit board **40**. In an exemplary embodiment of the present invention, the resistance value of the resistance element **190** on the first connector **10** is set to match the impedance of the circuit board **40**, so that the overall circuit balance can be achieved. For example, when the impedance of the circuit board **40** is 100 Ohms, the resistance value of each resistance element **190** of the first connector **10** is set equal to 100 Ohms. In the illustrated embodiment, when the first connector **10** is mated with the mating connector **50**, each pair of vacant terminals **120′** forms an electrical circuit with the circuit board **40** through the resistance element **190**, which can avoid the antenna effect of the vacant terminals **120′** and improve the reliability of signal transmission of the connector.
(27) It should be appreciated for those skilled in this art that the above embodiments are intended to be illustrative, and not restrictive. For example, many modifications may be made to the above embodiments by those skilled in this art, and various features described in different embodiments may be freely combined with each other without conflicting in configuration or principle.
(28) Although several exemplary embodiments have been shown and described, it would be appreciated by those skilled in the art that various changes or modifications may be made in these embodiments without departing from the principles and spirit of the disclosure, the scope of which is defined in the claims and their equivalents.
(29) As used herein, an element recited in the singular and preceded with the word “a” or “an” should be understood as not excluding plural of said elements or steps, unless such exclusion is explicitly stated. Furthermore, references to “one embodiment” of the present invention are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, embodiments “comprising” or “having” an element or a plurality of elements having a particular property may include additional such elements not having that property.

## Claims

1. A connector, comprising: an insulator; a plurality of pairs of terminals provided on the insulator, the plurality of pairs of terminals include at least one pair of cable terminals electrically connected with a cable and at least one pair of vacant terminals that are unused; and a resistance element electrically connected between each pair of vacant terminals.
2. The connector of claim 1, wherein the at least one pair of vacant terminals is two pairs of vacant terminals and the resistance element is one of a pair of resistance elements, each resistance element is connected in series between a pair of rear ends of each pair of vacant terminals.
3. The connector of claim 1, wherein the resistance element has a pair of pins, a rear end of each vacant terminal of the pair of vacant terminals is crimped or welded to one of the pins, a front end of each vacant terminal of the pair of vacant terminals is adapted to mate with a mating terminal of a mating connector.
4. The connector of claim 1, wherein the at least one pair of cable terminals is two pairs of cable

terminals electrically connected to two pairs of core wires of the cable.
5. The connector of claim 1, wherein a rear end of each of the cable terminals is crimped onto and electrically connected with a core wire of the cable, a front end of each cable terminal of the pair of cable terminals is adapted to mate with a mating terminal of a mating connector.
6. The connector of claim 1, further comprising a housing sheathed on the insulator, a shield shell sheathed on the housing, and a sealing ring compressed between the housing and the shield shell to seal a gap between the housing and the shield shell.
7. The connector of claim 6, further comprising a conductive screw sleeve sleeved on the housing and the shield shell and capable of threadably connecting with a mating shield shell of a mating connector.
8. The connector of claim 7, further comprising a conductive spring compressed between the shield shell and the conductive screw sleeve, the conductive spring electrically connecting the shield shell and the conductive screw sleeve.
9. The connector of claim 6, further comprising a heat shrinkable tube having a first end heat shrunk onto the shield shell and a second end heat shrunk onto the cable, a shielding layer of the cable is clamped between the heat shrinkable tube and the shield shell and welded to the shield shell via a copper foil.
10. The connector of claim 6, further comprising a sealant filled in a rear port of the shield shell and sealing the rear port.
11. The connector of claim 10, wherein a core wire of the cable extends into the shield shell from the rear port, the resistance element is arranged in the rear port.
12. The connector of claim 11, further comprising an isolation sleeve sleeved on the resistance element and separating the resistance element from the sealant.
13. The connector of claim 6, further comprising a plurality of positioning end caps assembled on a front end of the insulator and positioning a front end of each of the plurality of pairs of terminals, a pair of positioning holes are formed in each positioning end cap, the front ends of each of the plurality of pairs of terminals pass through the pair of positioning holes of the corresponding positioning end cap.
14. A cable connection assembly, comprising: a cable having a core wire and a shielding layer; and a first connector including an insulator, a plurality of pairs of terminals provided on the insulator, the plurality of pairs of terminals include at least one pair of cable terminals and at least one pair of vacant terminals that are unused, and a resistance element electrically connected between each pair of vacant terminals, the first connector is connected to a first end of the cable, the core wire of the cable is electrically connected to one of the cable terminals of the first connector, and the shielding layer of the cable is electrically connected to a shield shell of the first connector.
15. The cable connection assembly of claim 14, further comprising a tail sleeve that is an insert injection molded piece formed on a heat shrinkable tube of the first connector and the cable.
16. The cable connection assembly of claim 14, further comprising a second connector connected to a second end of the cable, the second connector includes a cable terminal electrically connected to the core wire of the cable, the cable terminal of the first connector is electrically connected to the cable terminal of the second connector via the cable.
17. A connector assembly, comprising: a cable connecting assembly including a cable and a connector connected to an end of the cable, the cable has a core wire and a shielding layer, the connector has an insulator, a plurality of pairs of terminals provided on the insulator, the plurality of pairs of terminals include at least one pair of cable terminals and at least one pair of vacant terminals that are unused, and a resistance element electrically connected between each pair of vacant terminals, the core wire of the cable is electrically connected to one of the cable terminals, and the shielding layer of the cable is electrically connected to a shield shell of the connector; and a mating connector matable with the connector.
18. The connector assembly of claim 17, wherein the mating connector includes a mating shield

shell and a plurality of pairs of mating terminals arranged in the mating shield shell, when the connector and the mating connector are mated together, the shield shell of the connector is electrically connected to the mating shield shell, and the plurality of pairs of terminals of the connector are mated with the plurality of pairs of mating terminals.

19. The connector assembly of claim 17, further comprising a circuit board, on which the mating connector is installed, when the connector is mated with the mating connector, each pair of vacant terminals forms an electrical circuit with the circuit board through the resistance element, a resistance value of the resistance element is set to match an impedance of the circuit board.

20. The connector assembly of claim 19, wherein the impedance of the circuit board and the resistance value of the resistance element is equal to 100 Ohms.

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